

Stony Brook University
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MIDTERM II PRACTICE PROBLEMS

Due: Optional, but you should use it to prepare for the midterm.

Instructions: This is an optional assignment whose purpose is to prepare you for the midterm. I strongly recommend that you try to solve or at least understand all the questions.

I. **Uncertainty: Insurance.** Consider the following scenario about choosing insurance coverage for transportation accidents.

- (1) Sheryl is very worried about car accidents next year. At the beginning of the year, Sheryl has \$10,000 in wealth, with no additional earnings for the year. With probability $2/3$ Sheryl has an accident and suffers a loss of \$7,500. (Sheryl is not hurt, just the car.) With probability $1/3$ there is no accident leading to zero loss. What is Sheryl's expected wealth?
- (2) From now on, assume that Sheryl's utility function over wealth is $u(w) = \sqrt{w}$, where w is Sheryl's final wealth. What is her expected utility?
- (3) Sheryl can purchase an insurance with a premium of \$5,100. This insurance will fully reimburse the damage (\$7,500) if an accident occurs and will give no payment in case of no accident. What is Sheryl's expected wealth if she takes the insurance? Is the insurance premium fair?
- (4) Will Sheryl take up the insurance? Compute the expected utility and compare to the utility in the second point.)
- (5) Alejandro is a friend of Sheryl. He hears Sheryl talk about her decision and he exclaims "I would not have purchased the insurance!" Give an example of a utility function such that Sheryl would *not* have purchased the insurance given a wealth of \$10,000 and the accident probabilities given above.
- (6) Alejandro adds: "Sheryl, you are so risk-averse, relax!" Leaving aside the 'relaxing' issue, provide intuition on why risk-aversion translates into a concave utility function, like $u(w) = \sqrt{w}$.

II. **Labor and time.** Mr. Pusheen lives for two periods, say when he is young and when he is old. Let $t = 1$ denote the period when he is young and $t = 2$ denote the period when he is old. In period $t = 1$, Mr. Pusheen is endowed with total hours $\bar{R} = 175,000$ which he can use between working and relaxing. Working is the only source of income of Mr. Pusheen when young. Mr. Pusheen receives an hourly wage rate of w period $t = 1$. On the other hand, Mr. Pusheen expects to retire when he is old, in period $t = 2$, which means that he will not work in this period, in fact, Pusheen only derives utility from consumption in period 2. Any income available for Mr. Pusheen in period $t = 2$ comes from savings in period 1. Mr. Pusheen can save part of his labor income earnings in period $t = 1$ at an interest rate of τ . Let c_1 and c_2 denote consumption in periods 1 and 2 respectively; R_1 and L_1 denote relaxation time and working time in period 1; finally s_1 denotes savings in period 1. Mr. Pusheen derives utility from consumption and relaxation as follows

$$U(c_1, c_2, R_1) = \sqrt{c_1} + \sqrt{R_1} + \delta \sqrt{c_2}.$$

Let the price of consumption in both periods be equal to 1.

- (1) Why are we assuming implicitly that Mr. Pusheen will not save anything in period 2?
- (2) State Mr. Pusheen's period 1 budget constraint.
- (3) State Mr. Pusheen's period 2 budget constraint.
- (4) Using the previous two items, use s_1 to consolidate the per-period budget constraints into the *intertemporal budget constraint*. Hint: The only variables that should appear in the intertemporal budget constraint are c_1, c_2 and R_1 .
- (5) State Mr. Pusheen's utility maximization problem.
- (6) Solve for the optimal levels of consumption and the optimal labor supply of Mr. Pusheen.

III. **Time and uncertainty.** Mr. Rilakkuma lives for two periods, $t = 1$ and $t = 2$. His only source of wealth is an amount W in period 1. Mr. Rilakkuma would like to consume pancakes: let c_1 and c_2 denote the quantities of pancakes he consumes in periods 1 and 2 respectively. Pancakes have a price of \$1 in both periods. Since Mr. Rilakkuma has no wealth in period 2, he will need to invest part of his initial wealth in a risky asset in period 1 that pays back at the beginning of period 2 to keep eating pancakes in this period. Let x denote the amount of money he invested in the risky asset in period 1. With probability π the asset has a high return of r_H and with probability $1 - \pi$ the asset has a low return of r_L . Both rates are strictly positive. If Mr. Rilakkuma invests an amount x in period 1, then he will receive $(1 + r^i)x$ where $i = H, L$, depending on whether the returns are high or low. Mr. Rilakkuma's utility from consumption is given by the Bernoulli utility function $u(c) = \log(c)$. Since the returns of the asset are subject to uncertainty, consumption in period 2 is *contingent on the returns*. Let c_2^H and c_2^L denote the state-contingent consumption when there are high returns and low returns respectively. Mr. Rilakkuma will therefore try to maximize his expected utility:

$$E[u] = \log(c_1) + \pi \log(c_2^H) + (1 - \pi) \log(c_2^L).$$

- (1) What is Mr. Rilakkuma's state-contingent consumption in the high return state?
- (2) What is Mr. Rilakkuma's state-contingent consumption in the low return state?
- (3) What is his budget constraint in period 1? (Hint: he can only consume or invest.)
- (4) Use the results from the previous three items to state Rilakkuma's expected utility maximization problem subject to the period 1 constraint in terms of choice variables c_1 and x .
- (5) Solve for the optimal levels of c_1 and x . Use the optimal level of x and items 1 and 2 to obtain the optimal state-contingent consumption plans in period 2.
- (6) How does the optimal levels of c_1, c_2^H, c_2^L and x vary with respect to changes to
 - The probability of a high-return π ?
 - The rates of return r^H and r^L ?

IV. **Profit maximization.** Pip owns a business that produces widgets y . In the short-run, producing widgets y is a labor-intensive operation. It is produced using labor with the production function $y = l^\alpha$, for all $\alpha > 0$. Pip pays his workers a wage rate of w . Pip maximizes his profits as follows:

$$\max_{l \geq 0} pl^\alpha - wl,$$

where p is the price of widgets.

- (1) Define the concept of returns to scale, that is, what do we mean by constant/increasing/decreasing returns to scale. Under what conditions for α the production of wine has decreasing returns to scale?
- (2) From now on, assume that $0 < \alpha < 1$. Obtain the first-order condition from Pip's profit maximization problem and interpret it.
- (3) Use the (2) to solve for Pip's optimal labor input demand and the optimal supply of widgets.
- (4) Here is a small twist: suppose now that widget production is risky. Specifically, suppose that the price of widgets, p fluctuates from month-to-month, and Pip needs to decide in advance how to produce and how many workers to hire before the price level is known. Suppose that the price p of widgets is 10 with probability ψ , and 5 with probability $1 - \psi$. Pip should maximize *expected profits*. State the maximization problem and explain why you are setting up the problem the way you are.
- (5) Derive Pip's first order condition for the problem in item (4) and interpret it.
- (6) Use item (5) to solve for Pip's optimal level of labor demand and optimal supply of widgets.
- (7) Suppose $\alpha = 1/2$. How do the optimal levels of l and y change with respect to the probability ψ ? Provide economic intuition for your answer.
- (8) Suppose instead that Pip knew that the price of wine was going to be for sure $\psi 10 + (1 - \psi) 5$, which is the expected value of the uncertain price. Using your solution for point 3, compute the quantity of labor hired by Pip in this case, and compare it to the quantity hired when the price is uncertain (the one from point 6). How do the two quantities compare? Discuss the reason for this result.